MTH205 Analysis in one variable

Tutorial 5 DATE: **14.02.2025**

(1) Show that every polynomial of odd degree with real coefficients has at least one real root.

(2) Let $f : [0,1] \to \mathbb{R}$ be a continuous function such that $f(0) = f(1)$. Prove that there exists a point $c \in [0, \frac{1}{2}]$ such that $f(c) = f(c + \frac{1}{2})$.

(3) Let $f : [a, b] \to \mathbb{R}$ be continuous function and assume that $f(a) < 0,\ f(b) > 0$. If we denote $s := \sup\left\{x \in [a, b] :\ f(x) < 0\right\}$ then $f(s) = 0$.

(4) Show that if $f,\ g$ are uniformly continuous on $A \subseteq \mathbb{R}$ then $f + g$ and $f \circ g$ are uniformly continuous on A.

(5) Verify uniform continuity of the following functions.

   (a) $f : [0, \infty) \to \mathbb{R}$ defined by $f(x) = x^2$, for $x \geq 0$.

   (b) $g : (0, \infty) \to \mathbb{R}$ defined by $g(x) = \sin(\frac{1}{x})$ for $x > 0$.

   (c) $h : \mathbb{R} \to \mathbb{R}$ defined by $h(x) = \frac{1}{1+x^2}$ for $x \in \mathbb{R}$.

(6) Let $f : A \to \mathbb{R}$ be a uniformly continuous function with $|f(x)| \geq M > 0$ for all $x \in A$ then show that $\frac{1}{f}$ is uniformly continuous.